

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9648/01

Paper 1 Multiple Choice

25 September 2015

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name and class on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

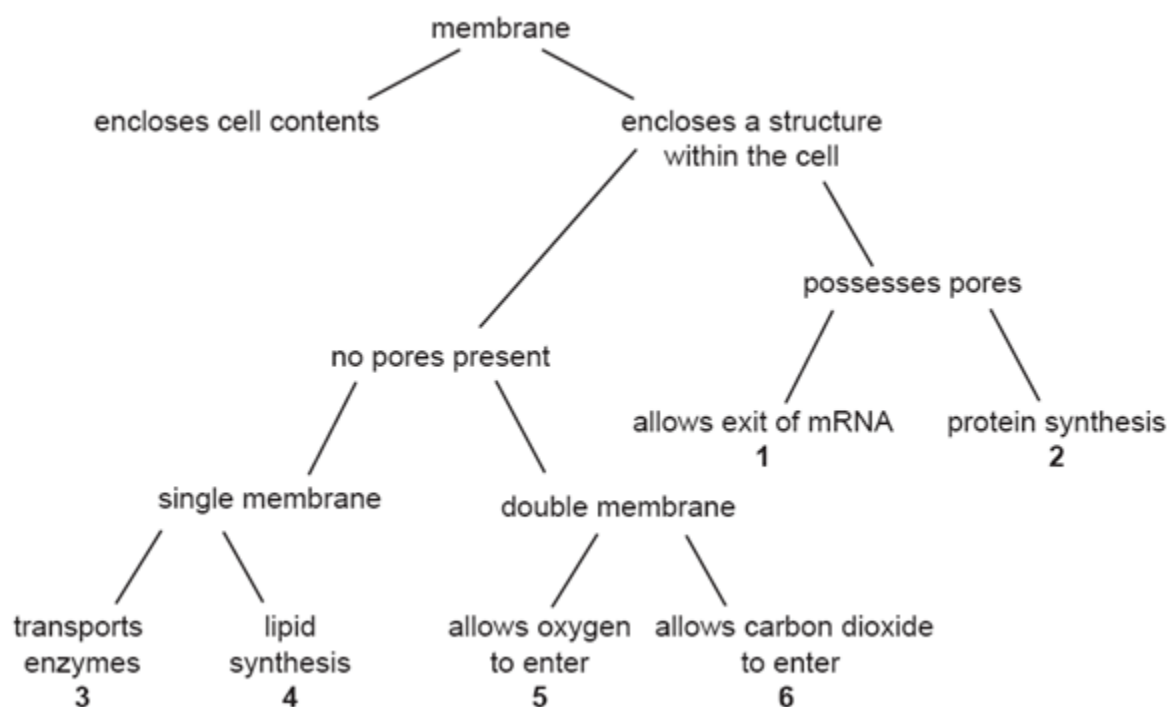
The use of an approved scientific calculator is expected, where appropriate.

This document consists of **14** printed pages and **0** blank pages.

[Turn over

1 Membranes within and at the surface of cells have different roles.

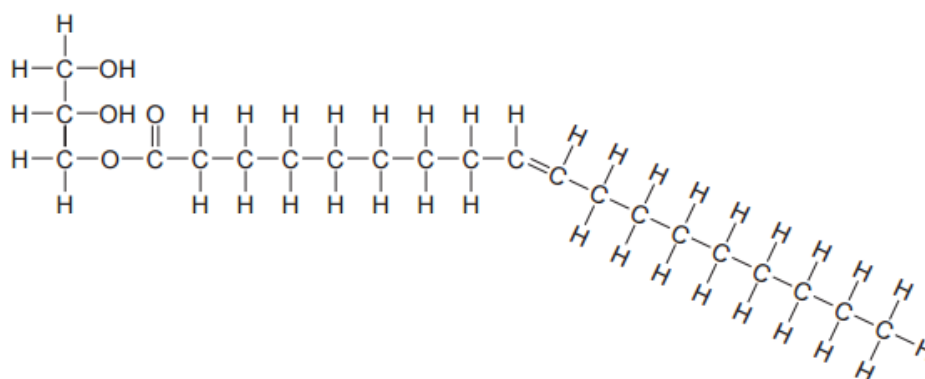
The diagram allows the identification of the various organelles within the cell, by describing the membrane structure and function.



Which of the outcomes shown below correctly identifies the organelles that possess the membrane and function concerned?

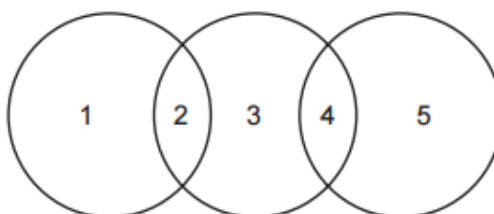
	1	2	3	4	5	6
A	nucleus	ribosome	vesicle	smooth ER	mitochondrion	chloroplast
B	nucleolus	rough ER	vesicle	smooth ER	nucleus	mitochondrion
C	nucleus	rough ER	vesicle	smooth ER	mitochondrion	chloroplast
D	nucleus	smooth ER	mitochondrion	rough ER	vesicle	chloroplast

- 2 The diagram shows a triglyceride molecule that has been partially hydrolysed.



What will be the products of the total hydrolysis of the molecule shown?

- A a molecule of glycerol and a saturated fatty acid molecule only
 - B a molecule of glycerol and an unsaturated fatty acid molecule only
 - C a molecule of water, a molecule of glycerol and a saturated fatty acid molecule
 - D a molecule of water, a molecule of glycerol and an unsaturated fatty acid molecule
- 3 The diagram shows the relationship between the levels of protein structure and bonds.



Which row is correct?

	1	2	3	4	5
A	primary	peptide	secondary	ionic	tertiary
B	secondary	hydrogen	tertiary	peptide	primary
C	tertiary	ionic	primary	peptide	quaternary
D	quaternary	ionic	tertiary	ionic	secondary

4 What is the effect of an enzyme in an enzyme-catalysed reaction?

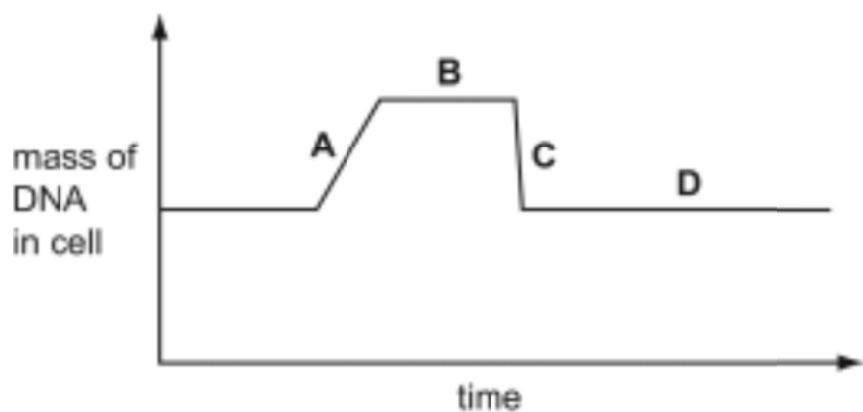
- A decreases the activation energy and has no effect on the energy yield
- B increases both the activation energy and the energy yield
- C decreases the activation energy and increases the energy yield
- D decreases both the activation energy and the energy yield

5 The electron micrograph shows a cell at a particular stage during the cell cycle.

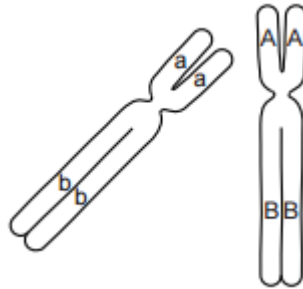


The graph shows the mass of DNA within the cell during the cell cycle.

When does the stage shown in the diagram occur?



- 6 The diagram shows two homologous chromosomes in early prophase I of meiosis in an animal cell. Two genes, A/a and B/b, whose loci occur on the homologous chromosomes are also shown.



Which row of diagrams is a possible representation of these chromosomes as they progress from anaphase I to prophase II?

	anaphase I	prophase II
A		
B		
C		
D		

- 7 The table shows the relative amounts of the bases adenine, thymine, guanine and cytosine in DNA from different organisms.

source	adenine	thymine	guanine	cytosine
bacterium	23.8	23.1	26.8	26.3
maize	26.8	27.2	22.8	23.2
fruit fly	30.7	29.5	19.6	20.2
chicken	28.0	28.4	22.0	21.6
human	29.3	30.0	20.7	20.0

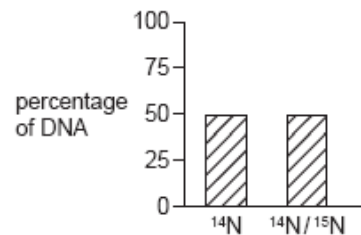
Which statements account for the importance of the ratios of A to T and G to C to the structure of DNA?

- 1 Complementary base pairing can occur.
- 2 Mutation will occur when pairing ratio is lost.
- 3 Semi-conservative DNA replication can occur to copy DNA strands.
- 4 Phosphodiester bonds helps to hold two strands together.
- 5 Purines and pyrimidines have different sizes and shapes.

- A** 1 and 3 only
- B** 1, 2, 3 and 5 only
- C** 2, 3, 4 and 5 only
- D** 1, 2, 3, 4 and 5

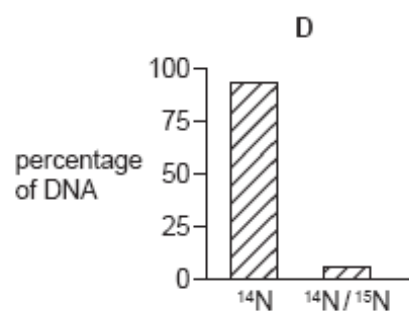
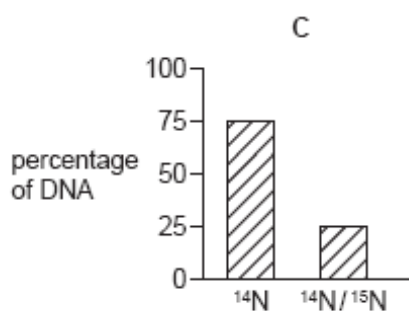
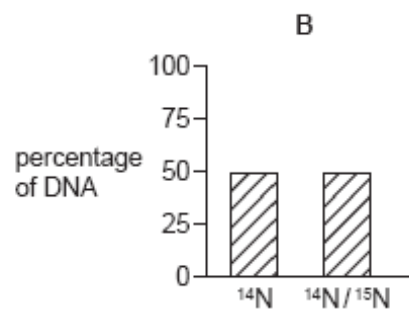
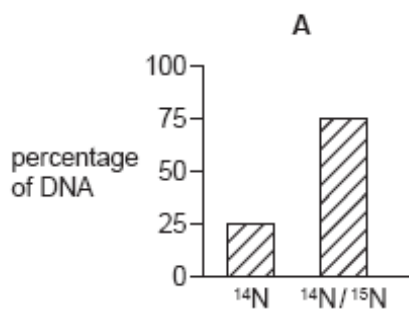
- 8 Bacteria were grown in a medium containing ^{15}N . After several generations, all of the DNA contained ^{15}N . Some of these bacteria were transferred to a medium containing the common isotope of nitrogen, ^{14}N . The bacteria were allowed to divide once. The DNA of some of these bacteria was extracted and analysed. This DNA was all hybrid DNA containing equal amounts of ^{14}N and ^{15}N .

Some bacteria from the medium with ^{15}N were transferred into a medium of ^{14}N . The bacteria were allowed to divide twice. The graph shows the percentages of ^{14}N and ^{15}N in the DNA of these bacteria.

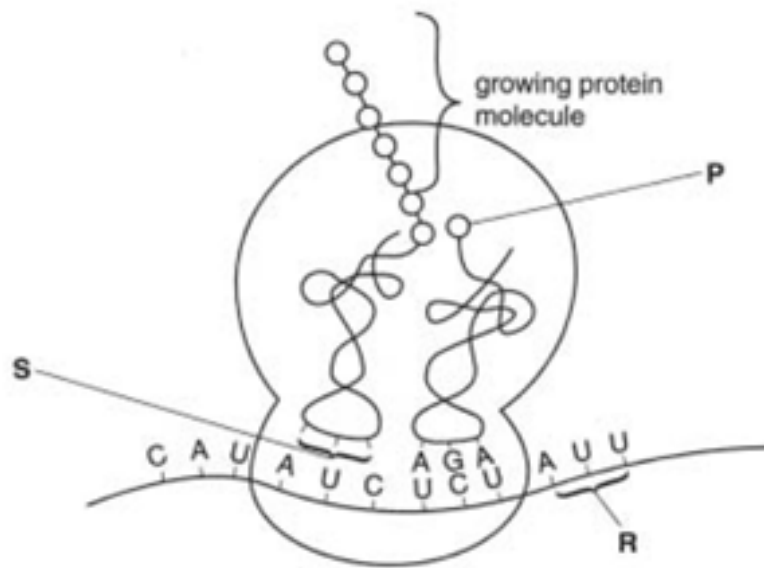


Some bacteria from the medium with ^{15}N were transferred into a medium of ^{14}N . The bacteria were allowed to divide three times.

What would be the percentages of ^{14}N and ^{15}N in the DNA extracted from these bacteria?



- 9 The diagram shows the process of translation occurring at a ribosome in a cell that synthesises secretory enzymes.



The table shows some triplet base sequences of mRNA and the amino acids for which they code.

mRNA	amino acid
AUU	isoleucine
AUC	isoleucine
AUG	methionine
AGA	arginine
UUA	leucine
UCU	serine
CAU	histidine

Which row correctly identifies P, S and R?

	P	S	R
A	arginine	UAG	codes for isoleucine
B	serine	GAU	codes for leucine
C	arginine	TAG	codes for leucine
D	serine	UAG	codes for isoleucine

10 In three different genetic dictionaries, the genetic code for the amino acid cysteine is given as:

ACA or ACG OR
TGT or TGC OR
UGU or UGC

The explanation for this may be:

- 1 Some genetic dictionaries show mRNA codons, others show DNA triplets.
- 2 Some genetic dictionaries show the triplet code complementary to the mRNA code, others show the triplet code for the other strand.
- 3 The genetic code can be read in either the 3' or 5' direction along the DNA.
- 4 The genetic code is a degenerate triplet code.

- A** 1 and 4 only
B 2 and 4 only
C 1, 2 and 3 only
D 1, 2 and 4 only

11 What is the common phenomenon in all viral infections?

- A** DNA replication
B mRNA synthesis
C glycoprotein synthesis
D RNA dependent-DNA synthesis

12 Which of these is a correct description of a form of genetic recombination in bacteria?

- 1 Crossing-over occurs between paired chromosomes in meiosis.
- 2 Conjugation occurs when a cell passes DNA to another cell by means of a sex pilus.
- 3 Transformation occurs when a bacteriophage carries a bit of DNA from a previous host cell to a new host cell.

- A** 2 only
B 3 only
C 2 and 3 only
D 1, 2 and 3

- 13** The diagram represents part of a length of DNA from a prokaryote operon.

The length of DNA is labelled W, X, Y and Z. They have different functions in the control of transcription of the operon.



What identifies the functions of W, X, Y and Z?

	W	X	Y	Z
A	promoter	operator	regulator	structural gene
B	promoter	regulator	operator	structural gene
C	regulator	promoter	operator	structural gene
D	structural gene	promoter	operator	regulator

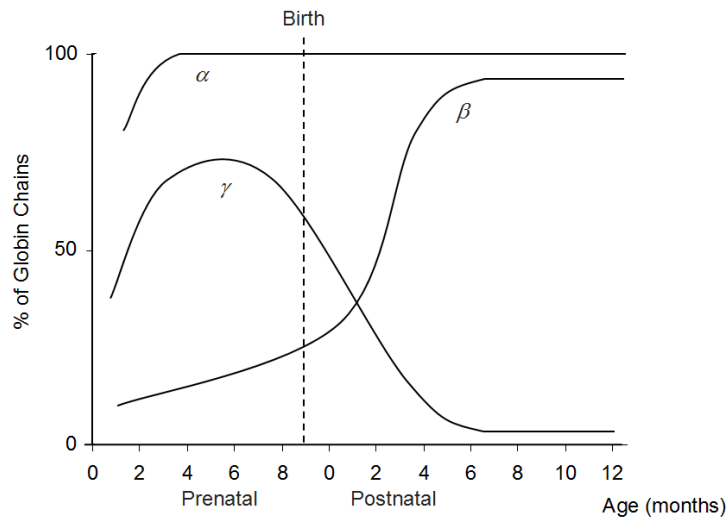
- 14** In a particular strain of *E. coli*, it was found that there was a change in DNA sequence of *lacI* gene. When grown in culture medium containing only lactose, it was found that this strain cannot utilise lactose as an energy source.

Which is a possible explanation for the observation above?

- A** β -galactosidase synthesised were unable to break down lactose into glucose and galactose.
 - B** Catabolite activator protein was not bound to CAP binding site, thus RNA polymerase was not recruited to the promoter.
 - C** *lac* repressor can no longer recognise and bind to the operator, thus the operon is turned off.
 - D** *lac* repressor remains active all the time as lactose cannot bind to its allosteric site.
- 15** Which statement applies to transcription in eukaryotic cells but not to prokaryotic cells?
- A** RNA polymerase transcribes the antisense strand of DNA to produce a strand of RNA.
 - B** During transcription, uracil replaces thymine in RNA.
 - C** Transcription takes place in the cell nucleus.
 - D** Initiation of transcription requires a promoter sequence of DNA.

- 16** The globin gene family in humans consists of α , β and γ genes. These genes code for the globin chains that make up haemoglobin and are expressed at different levels during different developmental stages.

The graph shows the expression of the various globin chains during the prenatal (fetal) and postnatal (after birth) periods.



Which statement cannot account for the differences in the levels of expression of globin chains?

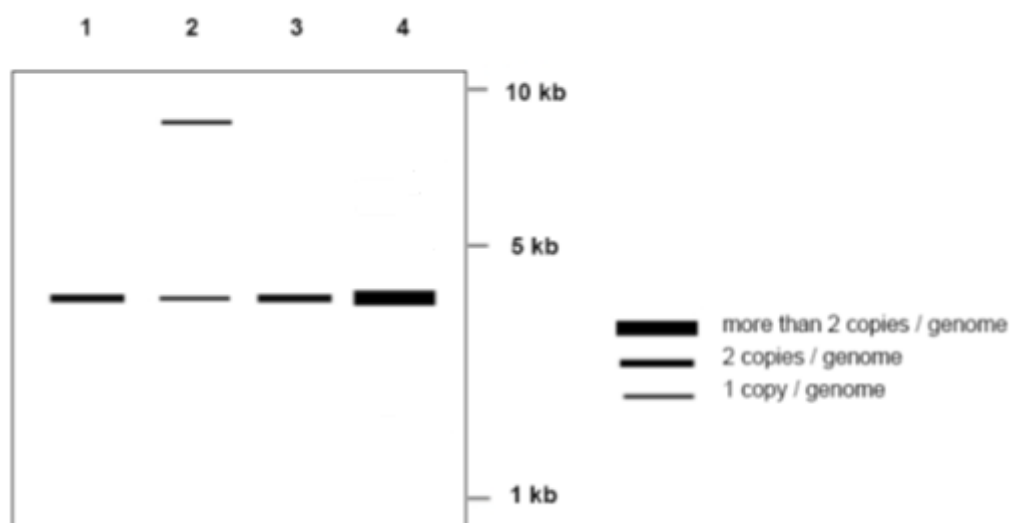
- A** Methyl groups are added to regulatory sequences of γ -globin genes during the prenatal period, allowing for some proteins to bind.
- B** A growth factor triggers the expression of a transcription factor that increases the rate of β -globin gene expression during the postnatal period.
- C** Alternative splicing occurs in the mRNA of the α -globin and β -globin genes, resulting in differences in the rate of expression of globin chains during the prenatal period.
- D** The shortening of poly(A) tail in the mRNA of γ -globin genes reduces its stability, resulting in a decrease in the rate of expression of γ -globin chains during the prenatal period.

- 17** Which statements about cancer are correct?

- 1 Exposure to UV light from the sun always causes skin cancer.
- 2 Some viruses can cause cancer.
- 3 The mutation of genes controlling the cell cycle may cause cancer.

- A** 1 and 2 only
- B** 1 and 3 only
- C** 2 and 3 only
- D** 1, 2 and 3

- 18 The *ras* oncogene is involved in a variety of human cancers. DNA isolated from a number of normal and cancerous human tissues were digested with *Eco*RI. A Southern blot was performed and the membrane was probed with radioactively-labelled cloned *ras* DNA. After exposure to x-ray film, the resulting autoradiograph is shown below.



lane	sample
1	white blood cells from a healthy human
2	lymphoma cells (cancerous)
3	bladder carcinoma cells (cancerous)
4	sarcoma cells (cancerous)

Based on the experimental results, which statement(s) could be deduced?

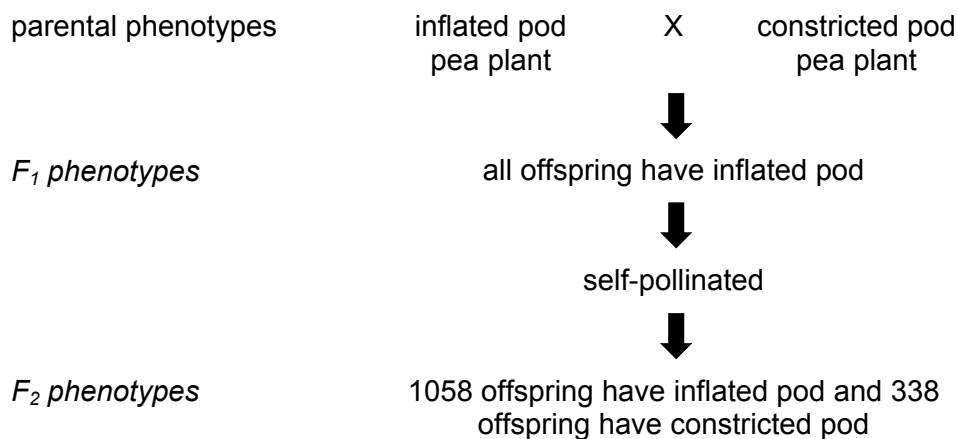
- 1 In tissue 3, there was no mutation within the *ras* coding region.
- 2 The result for tissue 2 could be explained by point mutations that caused the removal of an *Eco*RI site.
- 3 Gene amplification was involved in tissue 4.
- 4 In tissue 3, the mutation did not affect existing *Eco*RI sites within the *ras* gene.

- A 2 only
- B 3 and 4 only
- C 1, 2 and 4 only
- D 2, 3 and 4 only

19 A certain breed of mouse can have fur that is either black or brown. A mouse breeder performs a test cross using a brown mouse to determine the unknown genotype of a black mouse. There are four offspring produced by the cross. What conclusion can be drawn?

- A** If the black mouse is homozygous, one of the four offspring must be brown.
- B** If the black mouse is heterozygous, three of the four offspring must be black.
- C** The black mouse must be homozygous if all four offspring are black.
- D** The black mouse must be heterozygous if any of the offspring are brown.

20 In 1925, Sinnott and Dunn repeated one of Mendel's experiments and recorded the following result after crossing two pure-breeding pea plants.



What conclusions are consistent with these results?

- 1 Inflated or constricted pod shape in peas is dependent on the inheritance of a pair of codominant alleles for pod shape.
- 2 Each pea plant carries two alleles of the gene for pod shape in all its cells except for its gametes.
- 3 A gamete produced by a plant in the *F*₁ generation has an equal chance of carrying the allele for inflated pod shape or the allele for constricted pod shape.
- 4 The inflated pod plants in the *F*₂ generation all carry the allele for constricted pod shape.

- A** 1 and 2 only
- B** 1 and 4 only
- C** 2 and 3 only
- D** 2, 3 and 4 only

- 21** Skin colour in onions is controlled by two pairs of alleles, S/s and R/r, which segregate independently. The allele S is dominant and must be present to allow development of pigment in the skin. In its absence, the onion skin is white. Allele R is dominant and gives a red colour; the recessive r gives a yellow colour.

What will be the phenotypic ratio in the offspring of a cross between plants of genotype SsRR and ssrr?

- A** all red
- B** 1 red : 1 yellow
- C** 1 red : 1 white
- D** 1 white : 2 red : 1 yellow
- 22** What do chromosomal aberrations and gene mutations have in common and how are they different?

	similarity	difference
A	Both may involve addition of nucleotides.	Gene mutations always produce dominant alleles but not chromosomal aberrations.
B	Both may not result in disorders.	Gene mutations do not involve inversions but inversion of segments of chromosomes does occur.
C	Both affect DNA sequence.	Gene mutations occur within a chromosome but chromosomal aberrations may occur across chromosomes.
D	Both may not result in a difference in protein expression.	Chromosomal aberrations are more harmful than gene mutations.

23 Down's syndrome results from chromosomal aberration of chromosome 21.

This can occur in several ways.

- 1 During gamete formation the homologous pair of chromosomes of chromosome 21 failed to separate. A gamete with two copies of chromosome 21 is fertilized by a gamete with one copy of chromosome 21.
- 2 During early development of the embryo, chromatids of a chromosome 21 fail to separate.
- 3 Translocation of part of chromosome 21 onto chromosome 14 during the development of an embryo. During gamete formation chromosomes separate normally but may contain an extra chromosome 21.

Some individuals have a form of Down's syndrome called mosaicism. In these cases some of the body cells have the usual two copies of chromosome 21 and other cells have three copies of chromosome 21.

Which of the ways could explain how the mosaicism form of Down's syndrome occurs?

- A** 2 only
- B** 1 and 2 only
- C** 1 and 3 only
- D** 2 and 3 only

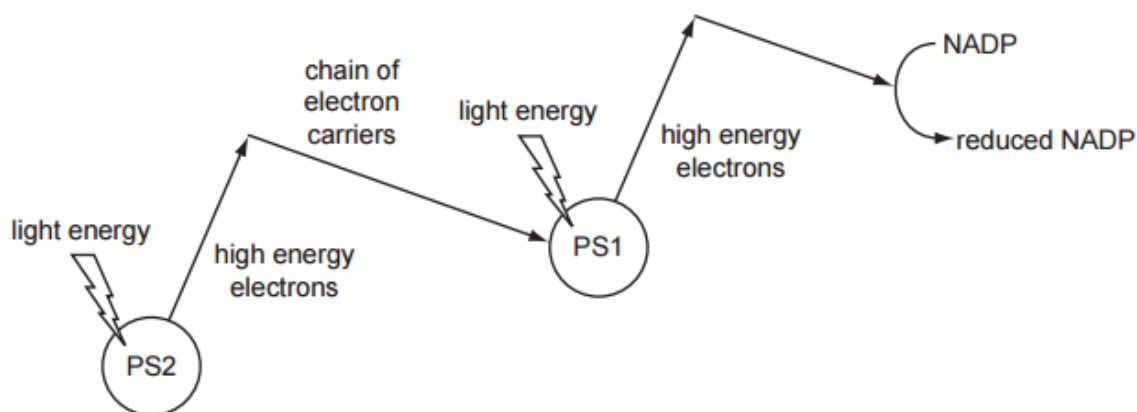
24 What causes variation in both sexually and asexually reproducing organisms?

- A mutations
- B polygenic inheritance
- C crossing over
- D independent assortment

25 If photosynthesising green algae are provided with CO_2 synthesised with heavy oxygen (^{18}O), later analysis will show that all but one of the following compounds produced by the algae contain the ^{18}O label. That one exception is

- A glycerate phosphate
- B glucose
- C RuBP
- D O_2

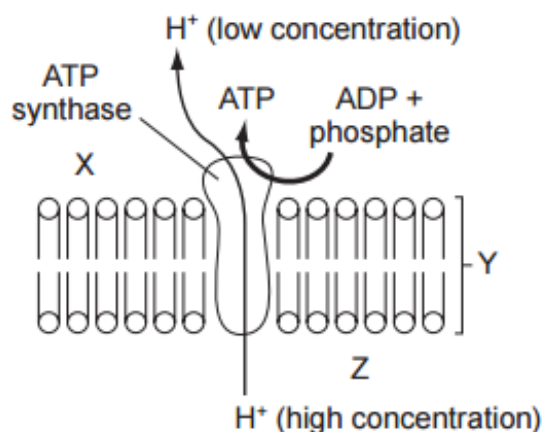
26 The diagram shows some of the processes in the light-dependent stage of photosynthesis.



For the light-dependent stage to continue, photosystem two (PS2) must gain electrons. Where do these electrons come from?

- A electron carriers
- B reduced NADP
- C photolysis
- D the formation of ATP

27 The diagram shows a membrane in a cell.



Which would be true of the diagram?

- A X is the thylakoid space, Y is the thylakoid membrane and the diagram shows ATP synthesis in a chloroplast.
- B X is the stroma, Y is the thylakoid membrane and the diagram shows ATP synthesis in a mitochondrion.
- C Y is the thylakoid membrane, Z is the cytosol and the diagram shows ATP synthesis in a chloroplast.
- D Z is the intermembranous space, X is the matrix and the diagram shows ATP synthesis in a mitochondrion.

28 The fluidity of the cell surface membrane can be changed by a number of factors.

As the fluidity of cell surface membranes decreases, which process would be least changed?

- A active transport
- B diffusion
- C endocytosis
- D osmosis

29 Which row correctly describes the different components of homeostatic mechanism for glucose regulation?

	stimulus	receptor	control centre	signal	effector
A	high glucose	islets of Langerhans	liver	glucagon	liver, muscles
B	high glucose	islets of Langerhans	islets of Langerhans	insulin	liver, muscles
C	lack of glucose	islets of Langerhans	islets of Langerhans	insulin	glucose
D	lack of glucose	liver	islets of Langerhans	glucagon	glucose

30 The table shows the descriptions of receptors used in cell signalling.

Which row shows a correct description for these receptors?

	receptor tyrosine kinase	G-protein-linked receptors	chemically-gated sodium channel receptors
A	changes shape when ligand binds	activates relay proteins	carries out autophosphorylation
B	activates relay proteins	carries out autophosphorylation	changes shape when ligand binds
C	carries out autophosphorylation	activates relay proteins	changes shape when ligand binds
D	carries out autophosphorylation	changes shape when ligand binds	activates relay proteins

31 Which is a correct description of the role of calcium ions in the neuromuscular system?

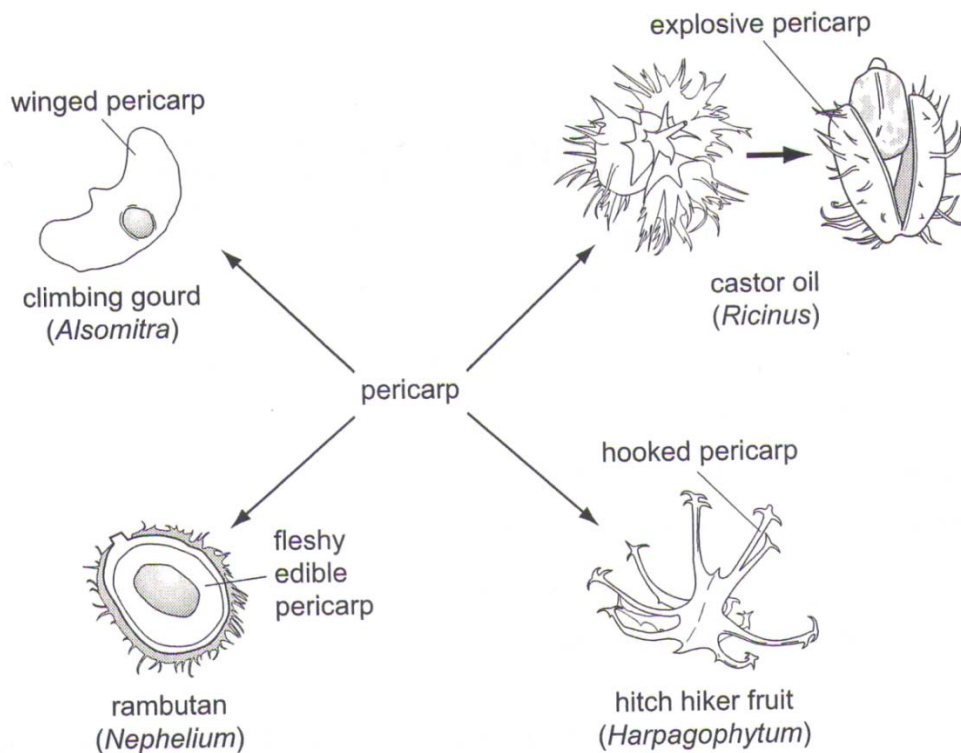
- A** exchanged with sodium ions through co-transport channels at axon surfaces during the reestablishment of a resting potential after an action potential
- B** moved in by diffusion through gated ion channels in pre-synaptic membranes of excitatory neurones causing vesicles to move to pre-synaptic membrane as an impulse arrives
- C** exchanged with chloride ions at the post-synaptic membrane, in changing membrane potential in inhibitory neurones
- D** actively pumped out of axons at nodes of Ranvier of myelinated neurones, being the main cause of the potential difference that is maintained during the resting potential

32 Which statements are correct interpretations of Darwinian evolutionary theory?

- 1 Advantageous behaviour acquired during the lifetime of an individual is likely to be inherited.
- 2 In competition for survival, the more aggressive animals are more likely to survive.
- 3 Species living in a stable environment will not evolve any further.
- 4 Variation between individuals of a species is essential for evolutionary change.

- A 4 only
- B 2 and 4 only
- C 1, 2 and 4 only
- D 1, 2, 3 and 4

33 The diagram illustrates variation in the pericarp (fruit wall) for a variety of methods of seed dispersal.



What do these examples illustrate?

- A The adaptive radiation of analogous structures showing convergent evolution.
- B The adaptive radiation of analogous structures showing divergent evolution.
- C The adaptive radiation of homologous structures showing convergent evolution.
- D The adaptive radiation of homologous structures showing divergent evolution.

34 The following statements relate to molecular phylogenetics.

- 1 Lines of descent from a common ancestor to present-day organisms have undergone similar, fixed rates of DNA mutation.
- 2 Organisms with similar base sequences in their DNA are closely related to each other.
- 3 The number of differences in the base sequences of DNA of different organisms can be used to construct evolutionary trees.
- 4 The proportional rate of fixation of mutations in one gene relative to the rate of fixation of mutations in other genes stays the same in any given line of descent.

Which statements, when taken together, suggest the existence of a 'molecular clock' that enables scientists to estimate the time at which one species might have diverged from another?

- A 1 and 2
B 1 and 4
C 2 and 3
D 3 and 4

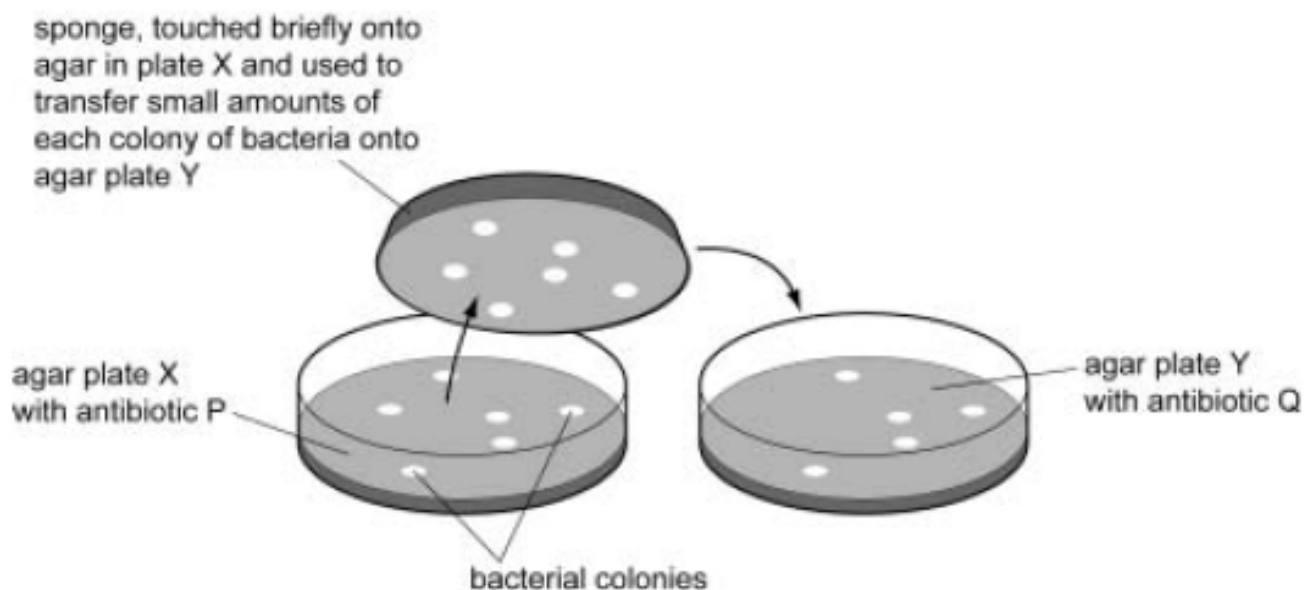
35 A piece of DNA 20 kilobase pairs (kbp) long was digested using different restriction enzymes, *Bam*HI, *Eco*RI and *Pst*I. The results are shown in the table below.

	Restriction enzyme used			
	<i>Pst</i> I	<i>Bam</i> HI	<i>Eco</i> RI	<i>Bam</i> HI <i>Pst</i> I
Lengths of DNA fragments (kbp)	15	17	12	12
	5	3	8	5
				3

Which set of fragments would result if all three enzymes were used together?

- A 7, 5, 3, 2
B 9, 5, 3
C 12, 3, 2
D 8, 7, 3, 2

- 36 The diagram shows a method used to detect which bacteria have been successfully transformed during genetic engineering.



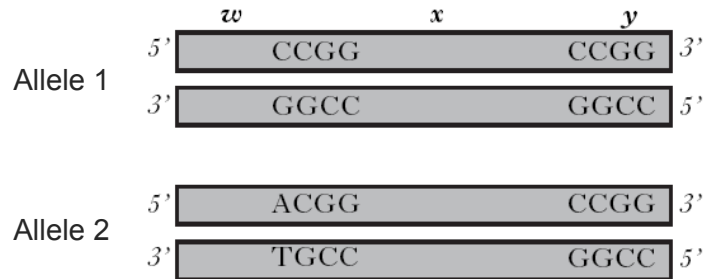
Which explains why other methods for detecting successful transformation are now preferred?

- 1 Incorporating heavy-metal resistance genes along with the desired genes means that you can easily kill cells that have not been transformed.
- 2 Presence or absence of non-toxic fluorescent markers is easy to detect using ultra-violet light.
- 3 The antibiotic resistance genes previously used as markers might have escaped into the environment.
- 4 The antibiotic resistance genes previously used as markers killed the transformed cells so they were difficult to use.

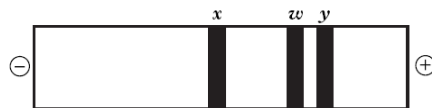
- A 1 and 3 only
- B 2 and 4 only
- C 1, 2 and 3 only
- D 1, 2, 3 and 4

- 37 A certain restriction enzyme will only cut a DNA strand between two Cs when the base sequence CCGG is present.

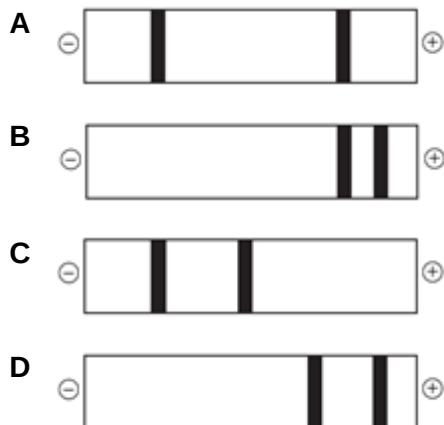
Two homologous segments of DNA that carry different alleles of a gene are shown below. Allele 2 has a single base-pair difference.



Allele 1 was treated with the restriction enzyme and the fragments w, x and y were obtained and separated by gel electrophoresis.



The resulting band pattern for allele 2 is



38 Some of the features of different types of stem cells are listed.

- 1 They are able to develop into all the cell types of the body to form a whole organism.
- 2 They can develop into a wide range of different types of cell.
- 3 They can divide indefinitely.
- 4 They can only develop into a limited range of cell types

Which feature(s) will not be shown by embryonic stem cells?

- A** 4 only
- B** 1 and 4
- C** 2 and 3
- D** 3 and 4

39 A patient underwent gene therapy using an adenoviral mediated delivery system. The gene therapy failed and the cause of the failure was attributed to the following reasons.

- 1 Failure of the expressed protein to be folded into the correct conformation during post-translational modifications.
- 2 The protein was not expressed in adequate amount.
- 3 The target gene failed to integrate next to a suitable promoter in the host chromosome.
- 4 Rejection and destruction of the vector by the host immune system.

Which explanations could be the reason behind the failure?

- A** 1 and 4 only
- B** 2, 3 and 4 only
- C** 1, 2 and 4 only
- D** 1, 2, 3 and 4

40 Some statements about genetically modified maize are listed.

- 1 Maize that is genetically modified to express Bt toxin is protected against several of its major insect pests.
- 2 The Bt toxin is effective against insect pests that eat the leaves and stems of the maize, but does not affect other animals that also eat the leaves and stems.
- 3 Some Bt maize expresses the gene in pollen that is wind-dispersed. This can dust the leaves of other species of plants and damage the insects feeding on them, such as Monarch butterfly caterpillars feeding on milkweed.
- 4 There is some evidence that the soil in which Bt maize is growing becomes toxic to various soil organisms during the growth of the Bt maize and for a short period afterwards.

Which of these statements might be used as an argument against planting a crop of maize that expresses the gene for Bt toxin only in its leaves, stems and roots?

- A** 4 only
- B** 1 and 2
- C** 2 and 3
- D** 3 and 4